

Abstract

Project Title

SmartNShine ATS Resume Generator with AI Interview Practice

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SmartNShine is a full-stack web application designed to assist job seekers in preparing effectively for the modern hiring process. The platform integrates three core capabilities — ATS-optimized resume building, AI-powered resume enhancement, and voice-enabled mock interview practice — into a single unified system built on the MERN stack (MongoDB, Express, React, Node.js).

Users can upload existing resumes or build new ones using a structured rich-text editor with multiple professional templates. The system leverages Google Gemini to parse, enhance, and score resume content, and analyzes resume-to-job-description alignment to provide actionable ATS feedback. For interview preparation, the platform uses OpenAI GPT-4o to generate contextual interview questions based on the user's resume or a target job role, evaluate spoken and typed answers across five dimensions — relevance, clarity, confidence, technical accuracy, and role fit — and produce a comprehensive post-interview performance report.

Voice interaction is supported through a Whisper-based speech-to-text microservice and a Chatterbox-based text-to-speech microservice, with browser-native fallback ensuring reliability across environments. The system includes a subscription and payment layer built on Razorpay, enforcing tiered feature access and AI usage quotas. An admin panel provides platform-wide analytics, user management, template control, and audit logging.

SmartNShine demonstrates how modern AI services, voice pipelines, and full-stack web development can be combined to address the end-to-end career preparation workflow in a single, accessible platform.

Keywords: ATS Resume, AI Interview Practice, MERN Stack, Google Gemini, OpenAI GPT-4o, Speech-to-Text, Text-to-Speech, Razorpay, Full-Stack Web Application